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Investigation of scaling relationships of lunar crater diameter-depth morphology in mare and highland terrains

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Abstract

Impact crater morphology on the Moon shows the interaction between impact energy, target composition and gravitational modification. This study investigates whether crater sizing on the moon follows a power-law relationship and also whether crater morphology differs between mare and highland terrain. A total of 591 craters were recorded from LunarQuickmap in a shared dataset among 15 students, 315 simple, 274 complex and 2 transitional craters were taken from both mare and highland regions. Diameter and depth of craters were taken from elevation profiles and plotted on to a log-log graph to test for power-law behaviour.

Results demonstrated a strong linear relationship between log diameter and log depth with a slope/scaling exponent of 0.3846. This value indicates that crater depth increases more slowly than diameter, this is consistent with the effect of gravitational modification in large craters. Simple craters displayed a stronger depth-diameter correlation than complex reflecting more excavation dominated processes rather than gravitational processes. Comparisons between mare and highland showed very similar scaling trends suggesting that impact energy is a more dominant factor in formation and morphology than target composition. However, subtle differences in trends were noticed among different terrains meaning it still has an effect. These results confirm past studies on crater morphology and also the transition of formation processes from strength/energy dominated to gravitational strength dominated as diameter increases.

Introduction

Impact craters are depressions on the surface of an astronomical body which usually have a circular or elliptical shape. These are formed by a high velocity impact from a smaller object which would normally originate from outside the body **(Lofgren et al.,1981).** On many solar system objects, impact craters are a dominant geographical feature such as the Moon, Mercury and Callisto. Craters on bodies experiencing active geological processes are less common since they become eroded by volcanic or tectonic processes over time, this can happen in places such as Earth, Venus, Titan and many more **(French, 1998)**. The features and characteristics of a crater can also be governed by many other factors such as the gravity and composition of the impact site.

The morphology of these features can be divided up into several categories as seen in Fig 1 below:

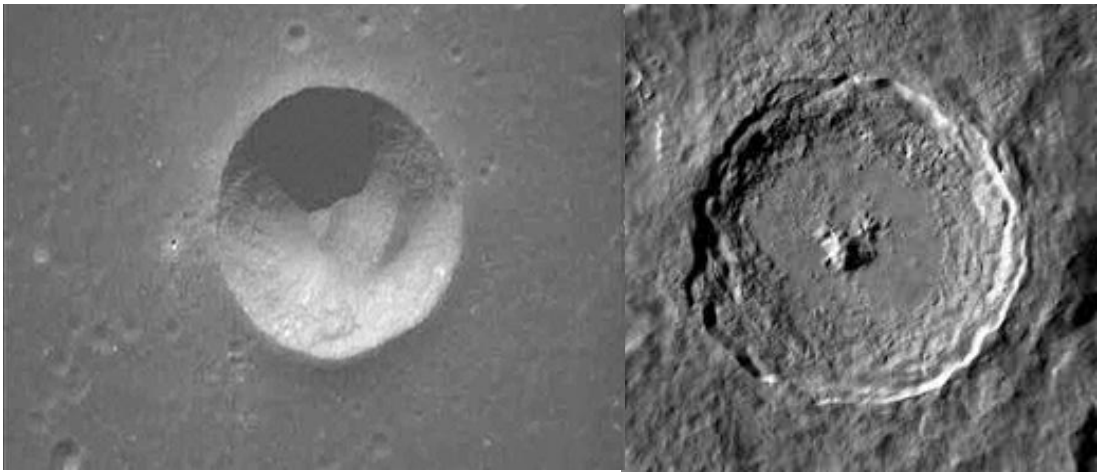


Figure 1 shows a simple crater (left) and a complex crater (right) ***(Tycho crater, 2026)***

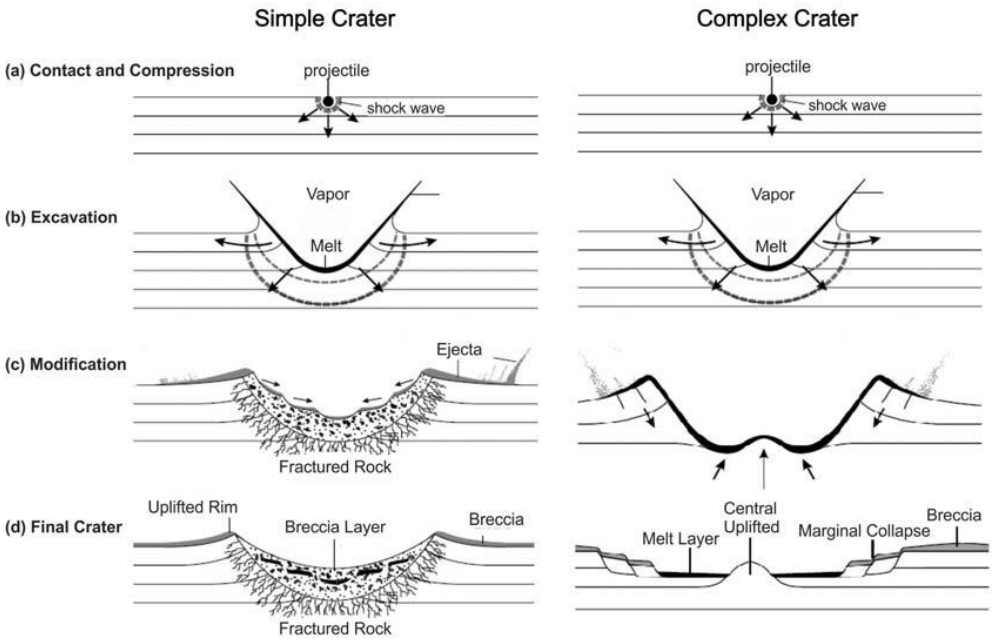


Figure 2 – A diagram showing a 2d view of the formation of simple and complex craters ***(Leite, 2015)***

Simple craters display a bowl-shaped basin whereas complex craters feature slumped walls and a central peak **(Melosh, 1989)**. Observations of both of these types of craters show that an increase in diameter of the crater forms a flatter floor and central topographical peak in the centre. The formation of this central peak allows us to differentiate between a complex and simple crater **(Osinski et al. 2019)**. Morphology of larger craters is affected by gravitational processes whereas the crater enlarges and steepens it becomes unable to sustain its shape under the influence of gravity **(Chandnani et al. 2019)**.

An important relationship in impact cratering is the power-law relationship where crater depth increases with crater diameter when raised to a power, stated in the formula below:

$$d=kD^n$$

d is depth, **k** is the target material properties, **D** is diameter and **n** is the scaling exponent

Diameter will not increase linearly with depth as seen in the equation above, it will increase proportionally when raised to a power. In order to help visualise this better, the values we collect for crater depth and diameter measurements will be marked on a log-log plot giving a linear trend, helping to compare different terrains and crater types.

Craters can be found all over the moon in 2 different terrain types. Mare are ancient dark volcanic plains made from cooled basaltic magma **(Lofgren, 1981)**; highlands are old brighter areas of crust made of ferroan anorthosite **(Chandnani et al. 2019, Krüger et al. 2018)** These two areas are compositionally different which may affect the morphology of a crater after impact.



Figure 3 – A picture of the full moon with mare and Highlands annotated ***(Moon 101, n.d.)***

With all these different variables, we are testing to see whether crater morphology and scaling relationships differ between mare and highland areas.

More specifically asking:

Does crater depth follow a power-law relationship?

Is crater morphology controlled mainly by impact energy, or does surface composition play a role too?

Results from this project will be compared to previous studies on lunar craters in order to strengthen confidence in the results.

Methodology

In order to measure the size of different individual craters, they cannot be directly measured on the moon. Instead for this project, a lunar database (Lunar Quickmap) will be used. This presents an entire 3d map of the moon with the ability to measure different sections using their digital drawing tools.

When researching Lunar crater depth and diameter, drawing tools can be used to create a line which touches both sides of the rim. The difference between the distances along the line drawn can be calculated using 'Distance along line(km)' on the right. This gives the diameter of the crater.

Lunar Quickmap also possesses the ability to show the topography of the surface where the line is placed, as seen in the figure above. The depth of the crater is then measured by taking

the difference between the highest and lowest point on the topography diagram. It is important that other qualitative and quantitative data is collected about the craters such as:

- Coordinates of crater – So crater can be relocated if remeasurement is required
- Whether the crater is simple or complex – without doing this we cannot determine differences between types of craters
- Geological setting of crater i.e. Mare or highland – This will help to learn whether the composition of the surrounding area affects crater size and morphology

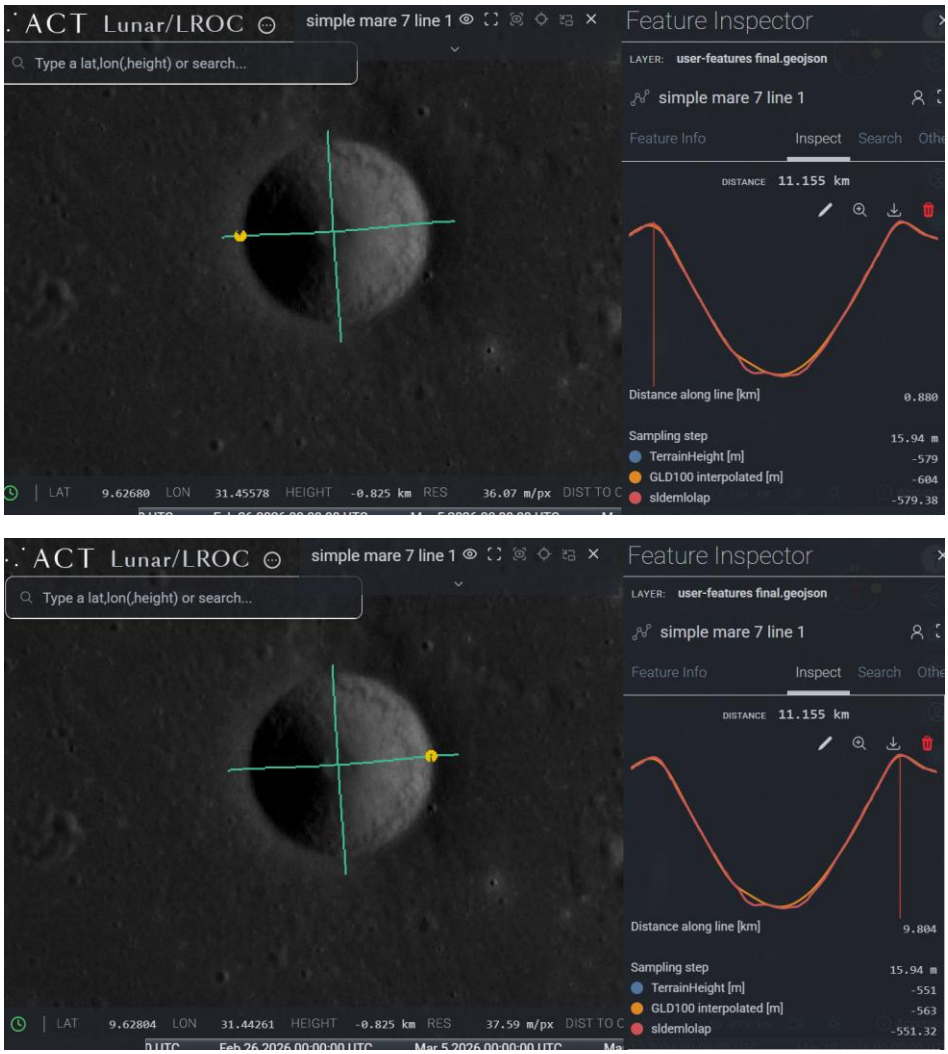


Fig 4 – screenshot of LunarQuickmap showing a selected crater on the moon's surface with the feature inspector on the right. This contains a red line profile of the surface's elevation. The yellow target can be moved along the line on the map which also shifts the vertical red line across the feature inspector. There are three different available elevation datasets (TerrainHeight, GLD100, sldemlolap) which can be used to calculate and depth of the crater. ***(Lunar and Planetary Institute, n.d.)***

The collected data can be inputted into Excel which has graphing and statistical analysis tools allowing for quantitative data to be presented and understood. We can plot this data on a log-log plot which will give us a straight-line $\log(d) = \log(k)+n\log(D)$.

If the scatter plot shows a clear linear trend, a power law relationship between crater depth and diameter can be confirmed.

Results

Summary of data collected

40 separate measurements of lunar craters were collected from the Lunar Quickmap dataset. This included 20 simple and 20 complex craters, 10 of each were from either mare or highland areas. To improve the final results and gain a more accurate statistical analysis, the collected data was combined with 15 other students including myself. This brought the dataset to a total of 591 individual craters. 315 were noted as simple, 274 were noted as complex and 2 were noted as transitional. Among the different area types, 300 were labelled to be found in Mare and 291 were labelled to be in the Highlands.

In the original raw data, there were two anomalous results that were identified due to the fact they did not follow the general trend and were out of proportion to the other values. This may have been due to a sampling or measurement error during data collection or manipulation. To avoid skewing the data, these points were removed.

Depth-Diameter relationship

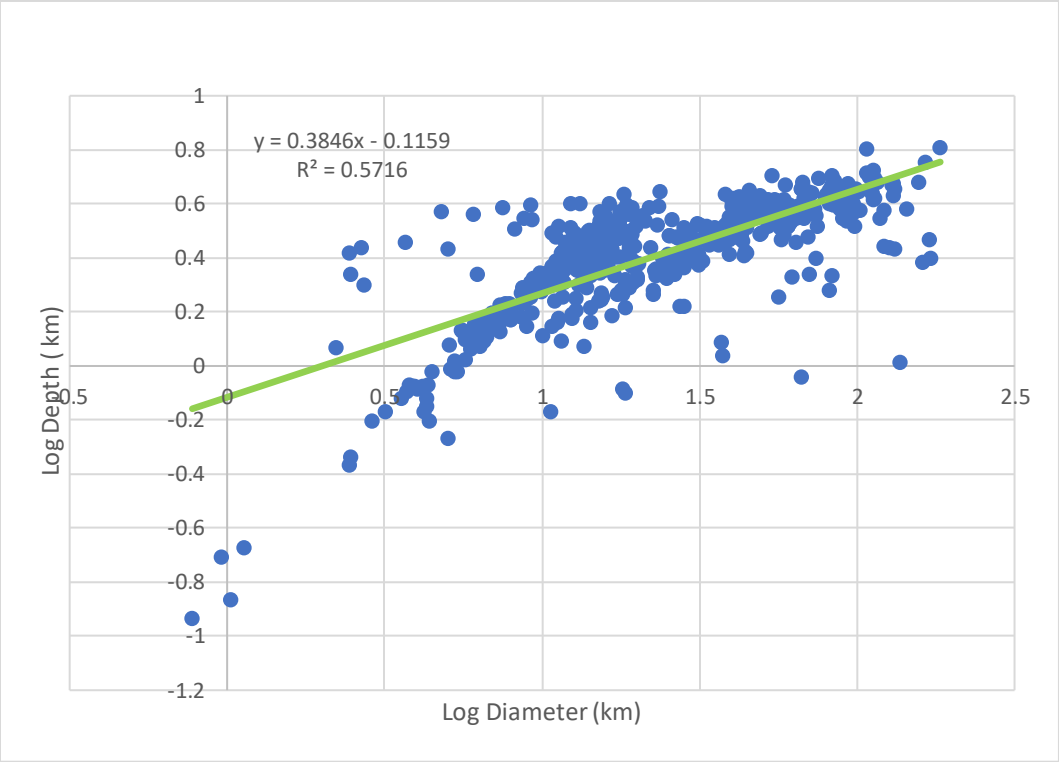


Figure 5 – a scatter graph for a total of 591 craters (blue) plotted log depth against log diameter with a trendline (green)

Figure 5 shows the relationship between log depth and log diameter. The graph clearly shows a positive linear correlation where increases in log depth = increases in log diameter. Furthermore, a linear regression calculation was performed giving a value of 0.3846 using the Excel data analysis feature. This number means that 1 unit increase in log Diameter means an increase of 0.3846 in log depth. The R^2 value (variation in depth explained by diameter) is 0.5716. A correlation test was done which gave a value of 0.75605.

Simple vs complex crater depth-diameter relationship

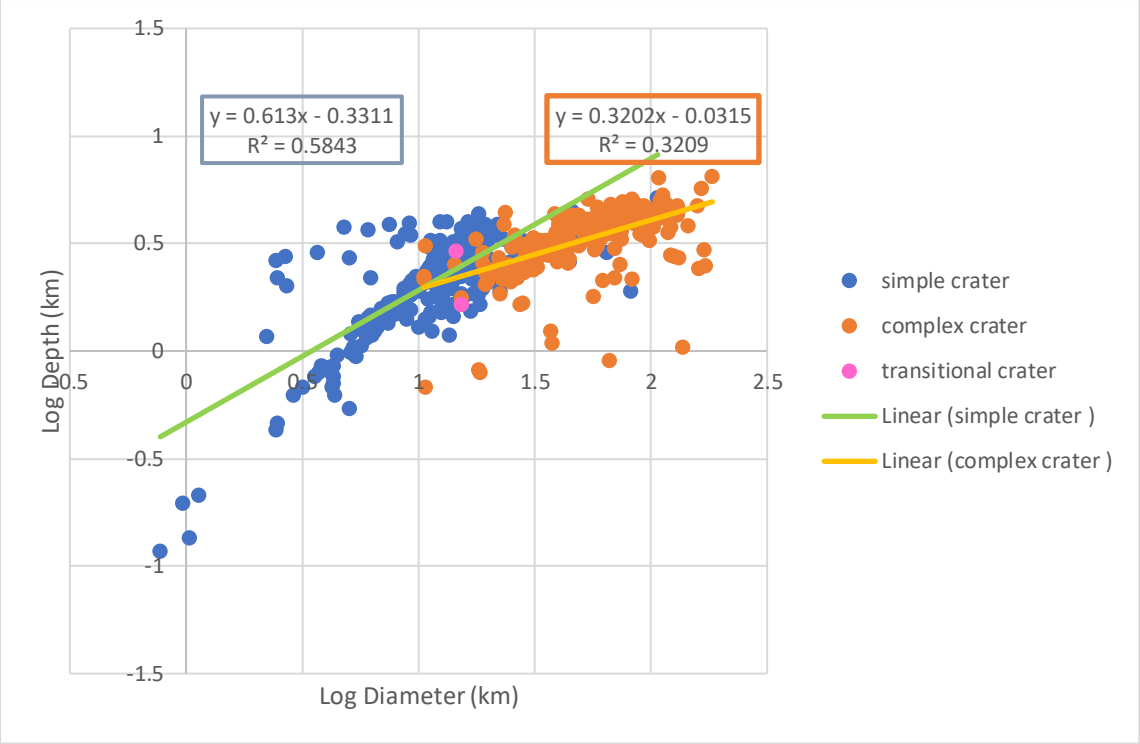


Figure 6– a scatter graph of total 591 craters which highlights simple (blue) and complex (orange) craters plotted log depth against log diameter

Figure 6 shows the diameter depth relationship between simple and complex craters. Immediately, it can be seen that complex craters on average are larger in depth and diameter than simple craters. The trendline for simple craters is steeper with a slope of 0.613 compared to complex which is 0.3202. Simple craters have a higher R^2 value (0.5843) than complex (0.3209). The point of divergence between the two types occurs when log Diameter = 1 or Diameter = 10km.

Mare vs highland depth-diameter relationship

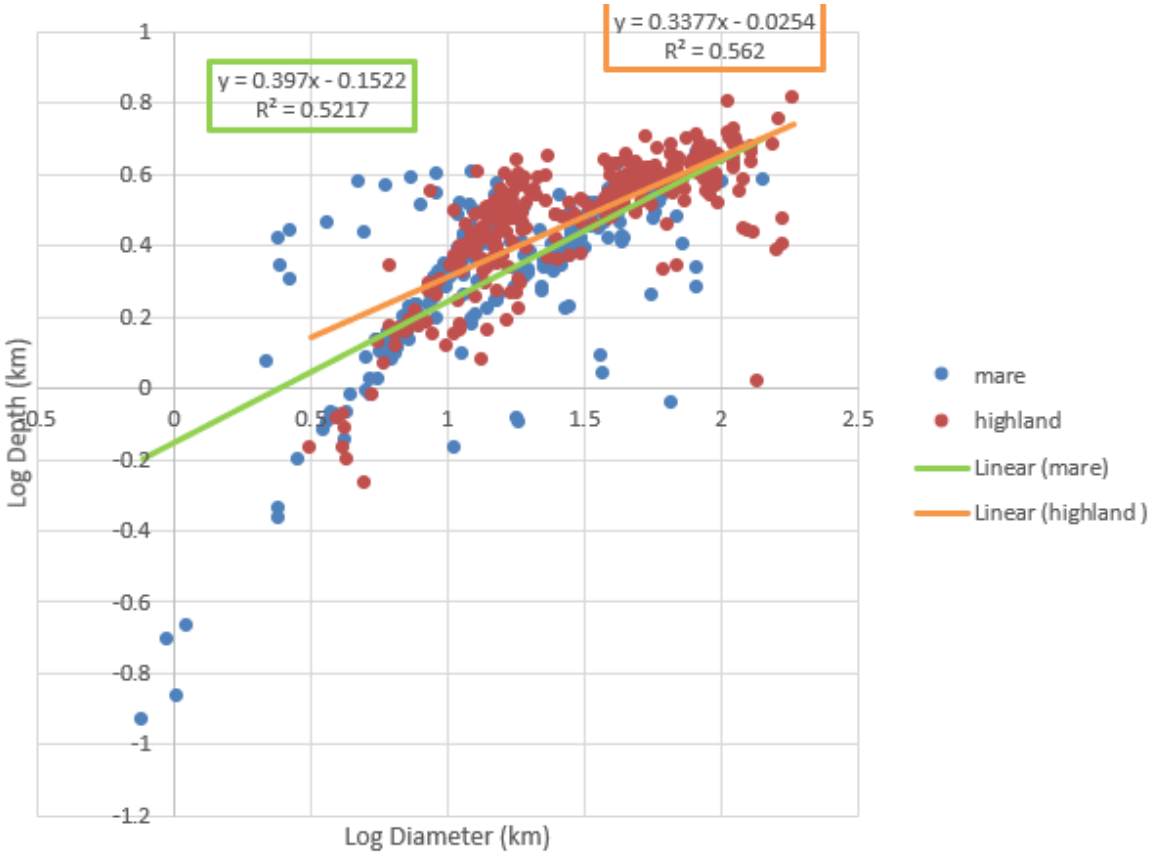


Figure 7 – a scatter graph showing simple + complex craters found in either mare or highland area (blue and red) with trendlines for each area (green and orange)

Figure 7 shows that both craters found in mare and highland follow a very similar trend in strong positive correlation (shown by trendline and slope of equation) and similar R^2 values (0.5217 and 0.562). Mare has a slightly steeper by 0.0593 meaning that depth of craters in mare increase faster than in highland craters.

My dataset vs global dataset

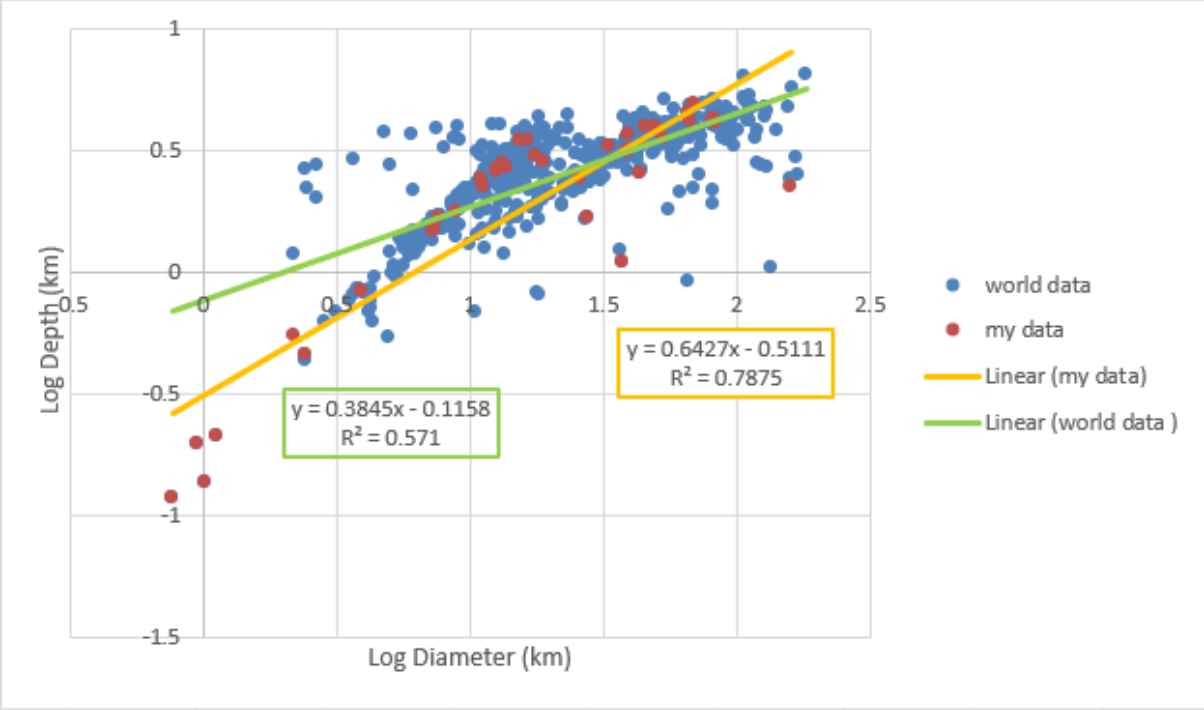


Figure 8 – a scatter plot graph showing my collected data plotted with the global dataset

In Figure 8 above, my dataset when compared to the global dataset shows a slightly differing relationship. Although they follow both a positive linear correlation, my dataset has a steeper trend and much higher R^2 value than the global dataset.

Discussion

From this project, we want to understand the factors which play a part in crater morphology and whether the relationship between the diameter and depth of the craters follow a clear power-law correlation.

Power law relationship

All of the log-log plot clearly shows a positive linear relationship following the equation stated earlier below:

$$\mathbf{d = kD^n}$$

Figure 3 shows all 591 craters plotted on a scatter graph showing the overall trend of the entire dataset. It gives a slope of 0.3846 indicating a positive correlation as for every 1km increase of log diameter is a 0.3846 km increase in depth for all craters on average. Depth increases slower on average than diameter, this reflects the effect of gravitational modification as crater size increases **(Melosh, 1989; Krüger et al., 2018).**

The results are similar to that seen in **Krüger et al. (2018)** where simple craters demonstrated a near perfect linear scaling relationship where the scaling exponent = 1 and complex craters showed a smaller exponent (0.3-0.4) which matches for the overall slope in Fig 3 for complex craters, suggesting that many of craters in my dataset are created under a gravitationally-dominated control.

Both a high R^2 value (0.5716) and correlation test number of 0.75605 confirms that a moderate amount of variability in crater depth is explained by crater diameter. This further supports the idea that impact energy is the main control on lunar crater morphology.

Simple-complex crater relationship

When plotting simple and complex craters separately on graph as shown in Figure 4, a clear difference can be seen in trends between both types. Firstly, it can be seen that simple craters hold a much higher R^2 value (0.5843) than complex craters (0.3209).

This pattern lines up with what was seen in **Krüger et al. (2018)** as simple crater depth linearly scales with diameter giving it a distinct bowl-shape profile, stated as an excavation dominated structure **(Krüger et al. 2018)**. On the other hand, complex crater morphology can be explained by gravitational collapse during the modification stage producing noticeable features such as:

- Slumped walls
- Central peak
- Terracing
- Floor uplift

The processes which form these features reduce the depth-diameter ratio giving a higher variability in the morphology of complex craters and increase in collapse-related processes. This is likely reflected by the lower R^2 value.

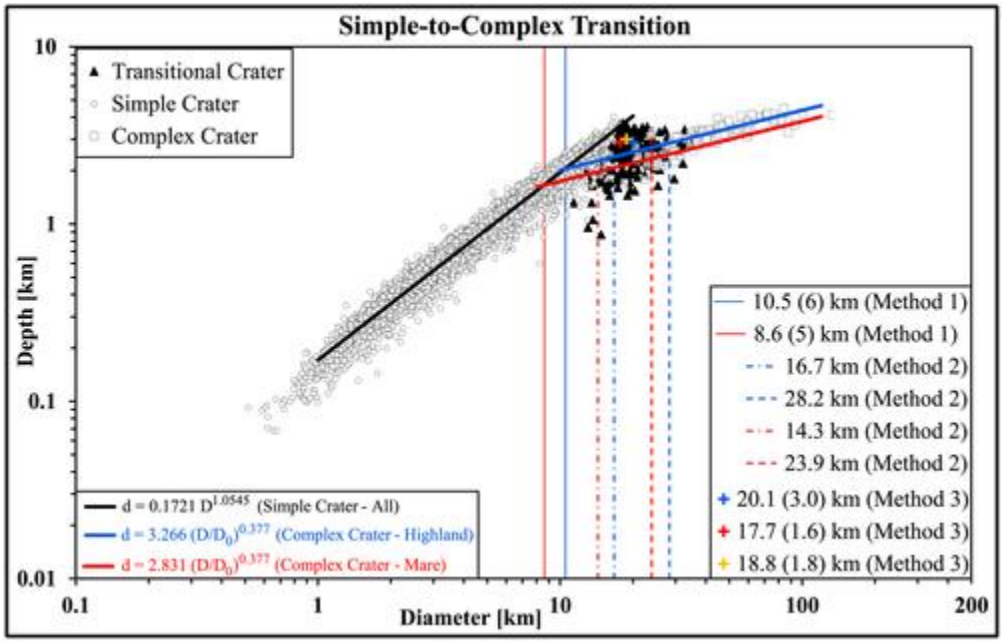


Figure 9 – Simple complex transition graph - **Krüger et al. (2018)**

Furthermore, Fig 9 shows the point of divergence between simple and complex craters occurs when log D = 1 (this is 10km in Diameter). This divergence of trendlines highlights the simple-complex transition point. However, this is slightly less in compared to the results in **Krüger et al. (2018)** where transition diameters occur over a range of diameters between 15-20km as seen in Fig 6. It is important to note though that the divergence occurs around 10km on Fig 6 but the presence of transitional craters occurs at higher diameters similar to fig 4 where the 2 transitional craters are present around Log diameter = 1.2 giving a diameter of 15.8km which aligns with the findings in **Krüger et al. (2018)**. This shows that the divergence of simple-complex trends may not be the best indicator of transition diameters. Due to lack of transitional data values, it is difficult to understand the range at which these occur, as transition is not abrupt but occurs across a range of diameters **(Osinski et al. 2019)**.

Mare-Highland comparison

In Figure 5, we can see the distinction between craters found in mare and highland terrain. In general, craters found in either area follow a similar behaviour with very little in difference between R^2 and slope values. Although in mare regions, the slope is slightly steeper compared to highland regions suggesting that depth increases much faster with diameter in basaltic regions of the moon.

According to **Krüger et al. (2018),** the trend of simple cratering is similar in both terrains but transitional diameters vary due to compositional factors within the surface.

The overall similarity between different terrains indicates impact energy to still be a dominant factor in crater morphology but the subtle differences in slope and R^2 values may reflect contrasts in influences in basaltic vs anorthosite compositions.

Therefore, crater morphology is controlled both by impact energy processes and target composition properties, with gravity being a more important factor as diameter increases.

Physical behaviour

The variation of slope/scaling exponent can explain a shift in physical processes which shape lunar crater morphology.

Smaller simple crater formation is dominated by target composition properties and excavation dominated processes.

More larger complex crater formation is dominated by collapse processes such as gravitational collapse.

As the overall size of the crater increases, gravity will exceed the strength of the terrain material causing features such as a central peak and slumped walls to form.

Conclusion

Limitations

It is important to acknowledge several limitations in this dataset:

- Transitional craters had very little representation with only 2 values out of 591 making it difficult to interpret
- Depth measurements were made from a single topographic profile which may not give the most accurate maximum depth
- Data is combined from 15 students leaving it prone to observer bias, inconsistent measurement technique and different classification thresholds
- Previous studies such as **Krueger et al. (2018), Osinski et al. (2019), chandnani et al. (2019)** used far larger data sets (up to thousands of individual measurements) than this report did which would have provided a more statistically accurate value for

slope and $R^{2.}$ However, there is still a consistency between this report's scaling exponent and other studies which verified visible trends in data.

Overall conclusion

This study confirms that lunar crater depth and diameter follow a power-law relationship. The primary control of crater sizing is impact energy. However, it is important to note that in larger craters that factors such as gravitational modification and surface composition can also influence the morphology. These results align with previous studies such as **Krueger et al. (2018)** and **Osinski et al. (2019)**, supporting the idea that morphology of craters transitions based on strength/energy processes to gravitational processes.

Appendix

	Latitude	Longitud	Average	Average	Complex	Mare/Hig	log diam	log depth (km)
1	3.41921	308.481	7.4245	1503.5	Simple	Mare	0.87067	3.1771
2	9.09548	305.242	7.716	1677.25	Simple	Mare	0.88739	3.2246
3	26.5175	20.2062	0.7755	117	Simple	Mare	-0.1104	2.06819
4	27.7449	11.7991	2.208	1175.37	Simple	Mare	0.344	3.07017
5	13.6071	60.7107	1.029	136	Simple	Mare	0.01242	2.13354
6	12.9035	62.6972	0.9615	196.25	Simple	Mare	-0.0171	2.29281
7	9.64691	30.9948	8.9025	1765.5	Simple	Mare	0.94951	3.24687
8	10.6696	25.1359	1.1275	213.25	Simple	Mare	0.05212	2.32889
9	7.02627	20.9597	2.455	459.25	Simple	Mare	0.39005	2.66205
10	6.91023	22.3954	3.943	837.25	Simple	Mare	0.59583	2.92286
11	-6.8569	8.99867	12.6105	2635.75	Simple	Highlanc	1.10073	3.4209
12	-5.0359	7.38307	13.7725	2697.25	Simple	Highlanc	1.13901	3.43092
13	-16.42	10.7476	13.4625	2796.5	Simple	Highlanc	1.12913	3.44661
14	27.253	50.6466	11.1865	2416.25	Simple	Highlanc	1.04869	3.38314
15	30.5786	47.3056	15.4455	2916.75	Simple	Highlanc	1.1888	3.4649
16	32.8488	46.0286	11.3225	2254.75	Simple	Highlanc	1.05394	3.3531
17	33.1585	106.213	7.3415	1459.5	Simple	Highlanc	0.86578	3.1642
18	37.3548	59.8365	16.599	2701.25	Simple	Highlanc	1.22008	3.43156
19	-13.431	88.203	12.7695	2595	Simple	Highlanc	1.10617	3.41414
20	-11.35	98.711	17.849	3015.5	Simple	Highlanc	1.25161	3.47936
21	26.7091	346.878	33.304	3273.75	complex	Mare	1.5225	3.51505
22	33.8629	1.19323	52.978	3559.5	complex	Mare	1.7241	3.55139
23	21.1075	293.315	43.9345	2550.25	complex	Mare	1.64281	3.40658
24	23.7103	312.483	39.8525	3501.75	complex	Mare	1.60046	3.54429
25	26.4289	290.82	37.491	1094.75	complex	Mare	1.57393	3.03931
26	53.3829	297.207	39.257	3126.25	complex	Mare	1.59392	3.49502
27	52.7014	316.505	39.8195	3174	complex	Mare	1.6001	3.50161
28	23.2409	330.83	25.9015	2412	complex	Mare	1.41332	3.38238
29	36.9226	286.822	28.0895	1669.5	complex	Mare	1.44854	3.22259
30	31.8207	292.289	18.968	2853.5	complex	Mare	1.27802	3.45538
31	44.1951	16.3898	67.5725	4156.75	complex	Highlanc	1.82977	3.61875
32	-32.539	54.1587	71.717	4355.5	complex	Highlanc	1.85562	3.63904
33	24.1827	79.1075	69.8755	4407.25	complex	Highlanc	1.84432	3.64417
34	-25.168	60.9075	161.322	2411.5	complex	Highlanc	2.20769	3.38229
35	-27.156	109.163	83.19	4209.5	complex	Highlanc	1.92007	3.62423
36	34.3999	56.6935	84.253	3952.75	complex	Highlanc	1.92559	3.5969
37	45.7062	88.6821	50.0695	3949.75	complex	Highlanc	1.69957	3.59657
38	43.6318	93.013	46.264	3944.5	complex	Highlanc	1.66524	3.59599
39	20.4713	117.981	82.8455	4233	complex	Highlanc	1.91827	3.62665
40	3.62569	133.065	66.605	4519	complex	Highlanc	1.82351	3.65504

	max_1	max_2	min	depth
1	-1567	-1545	-3033	1477
2	-1294	-1302	-3002	1704
3	-2610	-2608	-2730	121
4	-2.522	-2519	-3097	1836.24
5	-3770	-3752	-3901	140
6	-3521	-3518	-3716	196.5
7	-582	-551	-2278	1711.5
8	-1719	-1721	-1934	214
9	-1697	-1721	-2168	459
10	-1900	-1925	-2745	832.5
11	1452	1445	-1349	2797.5
12	505	1610	-1602	2659.5
13	690	909	-1835	2634.5
14	-433	-348	-2807	2416.5
15	509	343	-2684	3110
16	-1227	-802	-3165	2150.5
17	-2906	-2908	-4401	1494
18	-130	328	-2954	3053
19	-968	-1032	-3591	2591
20	457	343	-2749	3149
21	-1365	-1279	-4663	3341
22	-1367	-931	-4782	3633
23	-1560	-1751	-4065	2409.5
24	395	-370	-3277	3289.5
25	-764	-1071	-2035	1117.5
26	-1964	-1279	-4505	2883.5
27	-1615	-1743	-4734	3055
28	-461	-499	-2827	2347
29	-1549	-1176	-2854	1491.5
30	-1593	-1607	-4807	3207
31	-1310	259	-4644	4118.5
32	1435	1051	-3262	4505
33	22	2335	-3358	4536.5
34	517	222	-2301	2670.5
35	1121	941	-3045	4076
36	721	-179	-3967	4238
37	1601	1656	-1827	3455.5
38	1418	1714	-2772	4338
39	1278	160	-3537	4256
40	4510	4628	-151	4720

	max_1	max_2	min	depth
1	-1508	-1500	-3034	1530
2	-1359	-1326	-2993	1650.5
3	-1623	-2611	-2730	113
4	-2530	-2525	-3042	514.5
5	-3780	-3756	-3900	132
6	-3532	-3510	-3717	196
7	-461	-528	-2314	1819.5
8	-1720	-1719	-1932	212.5
9	-1717	-1696	-2166	459.5
10	-1915	-1895	-2747	842
11	935	1291	-1361	2474
12	1279	985	-1603	2735
13	939	1126	-1926	2958.5
14	-492	-294	-2809	2416
15	27	2	-2709	2723.5
16	-502	-1186	-3203	2359
17	-2976	-2938	-4382	1425
18	-1379	30	-3024	2349.5
19	-1087	-919	-3602	2599
20	642	562	-2280	2882
21	-1293	-1284	-4495	3206.5
22	-1086	-1116	-4587	3486
23	-1236	-1554	-4086	2691
24	-348	70	-3853	3714
25	-1028	-1258	-2215	1072
26	-1371	-901	-4505	3369
27	-1617	-1375	-4789	3293
28	-648	-264	-2933	2477
29	-1232	-1015	-2971	1847.5
30	-1644	-1376	-4010	2500
31	-773	187	-4488	4195
32	1251	689	-3236	4206
33	625	1859	-3036	4278
34	-328	-73	-2353	2152.5
35	1592	1286	-2904	4343
36	-581	66	-3925	3667.5
37	2070	2872	-1973	4444
38	739	1149	-2607	3551
39	1161	259	-3500	4210
40	2815	5293	-264	4318

	start	end	dif.	start	end	dif.
1	1.119	8.802	7.683	1.327	8.493	7.166
2	1.427	9.161	7.734	0.92	8.618	7.698
3	0.098	0.902	0.804	0.098	0.845	0.747
4	0.57	2.772	2.202	0.451	2.665	2.214
5	0.286	1.345	1.059	0.237	1.236	0.999
6	0.263	1.239	0.976	0.078	1.025	0.947
7	0.917	9.804	8.887	0.258	9.176	8.918
8	0.003	1.143	1.14	0.105	1.22	1.115
9	0.208	2.662	2.454	0.08	2.536	2.456
10	1.236	5.115	3.879	0.426	4.433	4.007
11	2.34	15.29	12.95	1.824	14.095	12.271
12	1.904	15.734	13.83	1.814	15.529	13.715
13	1.548	14.998	13.45	1.084	14.559	13.475
14	0.556	12.11	11.554	0.603	11.422	10.819
15	2.055	17.78	15.725	1.97	17.136	15.166
16	0.973	12.155	11.182	0.98	12.443	11.463
17	1.564	8.916	7.352	1.174	8.505	7.331
18	2.123	18.758	16.635	2.657	19.22	16.563
19	1.893	14.702	12.809	3.615	16.345	12.73
20	2.714	20.329	17.615	2.206	20.289	18.083
21	5.487	38.574	33.087	2.93	36.451	33.521
22	7.785	60.888	53.103	7.898	60.751	52.853
23	7.248	50.733	43.485	2.77	47.154	44.384
24	6.121	45.747	39.626	4.652	44.731	40.079
25	8.052	45.93	37.878	4.871	41.975	37.104
26	6.39	45.626	39.236	4.065	43.343	39.278
27	8.39	49.686	41.296	5.693	44.036	38.343
28	6.547	32.641	26.094	6.217	31.926	25.709
29	5.161	33.314	28.153	6.242	34.268	28.026
30	2.26	21.412	19.152	1.739	20.523	18.784
31	11.203	76.847	65.644	9.031	78.532	69.501
32	10.037	79.451	69.414	7.64	81.66	74.02
33	10.823	77.32	66.497	7.224	80.478	73.254
34	13.363	174.336	160.973	13.031	174.702	161.671
35	13.423	95.507	82.084	8.236	92.532	84.296
36	7.837	90.129	82.292	6.865	93.079	86.214
37	6.564	57.169	50.605	4.737	54.271	49.534
38	6.502	52.278	45.776	3.831	50.583	46.752
39	8.309	92.214	83.905	6.357	88.143	81.786
40	7.888	75.334	67.446	5.834	71.598	65.764

Figure 10 – a screenshot of my data collected from Lunar Quickmap

References

Chandnani, M., Herrick, R. R. and Kramer, G. Y. (2019). 'Geologic Investigation of Deep Simple Craters in the Lunar Simple-to-Complex Transition', *Journal of Geophysical Research: Planets*, 124 (10), pp. 2482-2504. https://doi.org/https://doi.org/10.1029/2018JE005903.

French, B. M. (1998). *Traces of catastrophe: A handbook of shock-metamorphic effects in terrestrial meteorite impact structures*.

Krüger, T., Hergarten, S. and Kenkmann, T. (2018). 'Deriving Morphometric Parameters and the Simple-to-Complex Transition Diameter From a High-Resolution, Global Database of Fresh Lunar Impact Craters (D ≥ ~ 3 km)', *Journal of Geophysical Research: Planets*, 123 (10), pp. 2667-2690. https://doi.org/https://doi.org/10.1029/2018JE005545.

Lofgren, G. E. B., A. E.; Duke, Michael B.; Dungan, Michael A.; Green, John C.; Haggerty, Stephen E.; Haskin, L.A. (1981). 'Basaltic Volcanism on the Terrestrial Planets', p. p.765.

Melosh, H. J. (1989). 'Impact cratering: A geologic process', *New York: Oxford University Press; Oxford: Clarendon Press*.

Osinski, G. R., et al. (2019). 'Transitional impact craters on the Moon: Insight into the effect of target lithology on the impact cratering process', *Meteoritics & Planetary Science*, 54 (3), pp. 573-591. https://doi.org/https://doi.org/10.1111/maps.13226.

Figures

Figure 1 - The Open University (no date) *Figure _unit3.3.11 (Anville: simple bowl-shaped; 11 km diameter)* [online image], *OpenLearn: Earth's Resources and Systems – Craters*. Available at: https://www.open.edu/openlearn/mod/oucontent/view.php?id=95866&section=_unit3.3.10 (Accessed: 5th November 2025).

Tycho (lunar crater) (2025) *Wikipedia — The Free Encyclopedia* [online]. Available at: https://en.wikipedia.org/wiki/Tycho_(lunar_crater) (Accessed:5th November 2025).

Figure 2 - Leite, E. P. (2015) *Stages of simple and complex craters formation: (a) Contact and compression of the target rock; (b) Transient crater excavation and disintegration of the projectile; (c) Fracturing and central uplifting; (d) Final shape of a simple and complex crater (French, 1998)* [online image], *3D gravity modelling of impact structures in basaltic formations in Brazil: Part I – Vargeão, Santa Catarina*. ResearchGate. Available at: https://www.researchgate.net/figure/Stages-of-simple-and-complex-craters-formation-a-Contact-and-compression-of-the-target_fig3_281108413 (Accessed:5th November 2026).

Figure 3 - *Moon 101* (n.d.) *Moon 101* [PDF], Walthill Public School in collaboration with the *Lunar and Planetary Institute*. Available at: https://www.lpi.usra.edu/exploration/education/hsResearch/presentations/2016-2017/walthill.pdf (Accessed:15th February 2026).

Figure 4 - Lunar and Planetary Institute (n.d.) *Lunar Quickmap showing crater* [screenshot], Available at: https://quickmap.lroc.asu.edu (Accessed: 15th February 2026).

Figure 5 - a scatter graph for a total of 591 craters (blue) plotted log depth against log diameter with a trendline (green)

Figure 6 - a scatter graph of total 591 craters which highlights simple (blue) and complex (orange) craters plotted log depth against log diameter

Figure 7 - a scatter graph showing simple + complex craters found in either mare or highland area (blue and red) with trendlines for each area (green and orange)

Figure 8 - a scatter plot graph showing my collected data plotted with the global dataset

Figure 9 – simple-complex transition diagram

Krüger, T., Hergarten, S. and Kenkmann, T. (2018). 'Deriving Morphometric Parameters and the Simple-to-Complex Transition Diameter From a High-Resolution, Global Database of Fresh Lunar Impact Craters (D ≥ ~ 3 km)', *Journal of Geophysical Research: Planets*, 123 (10), pp. 2667-2690. https://doi.org/https://doi.org/10.1029/2018JE005545.

Figure 10 – Table of my colllected data